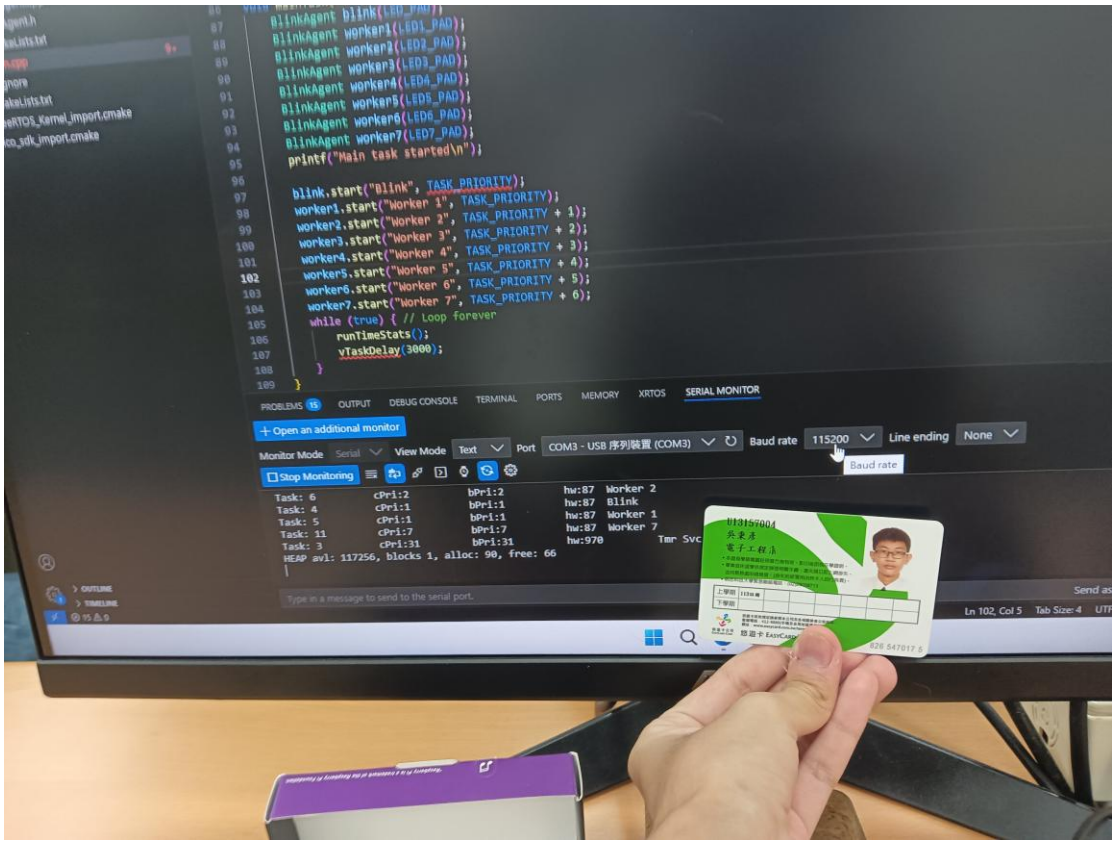
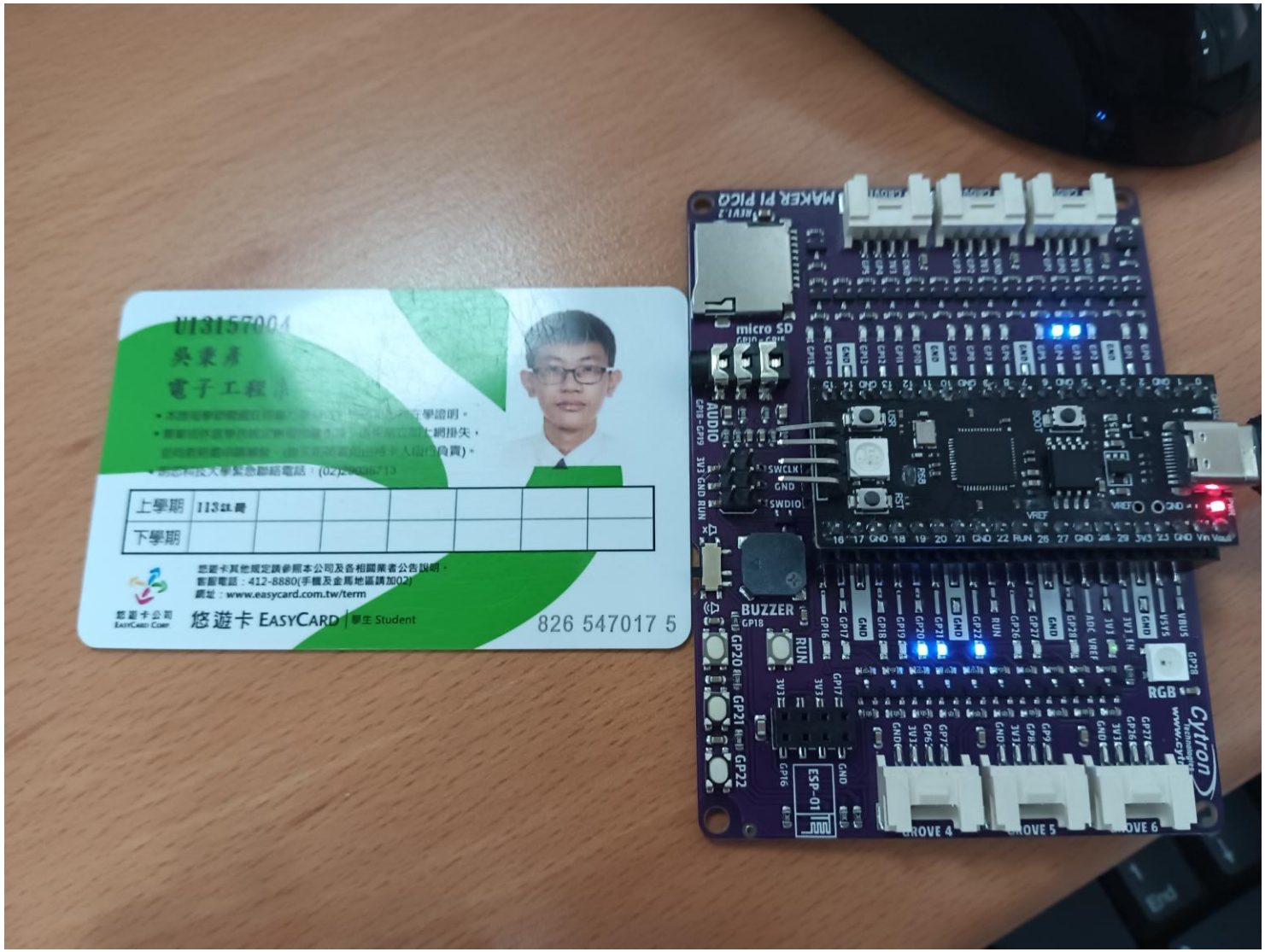
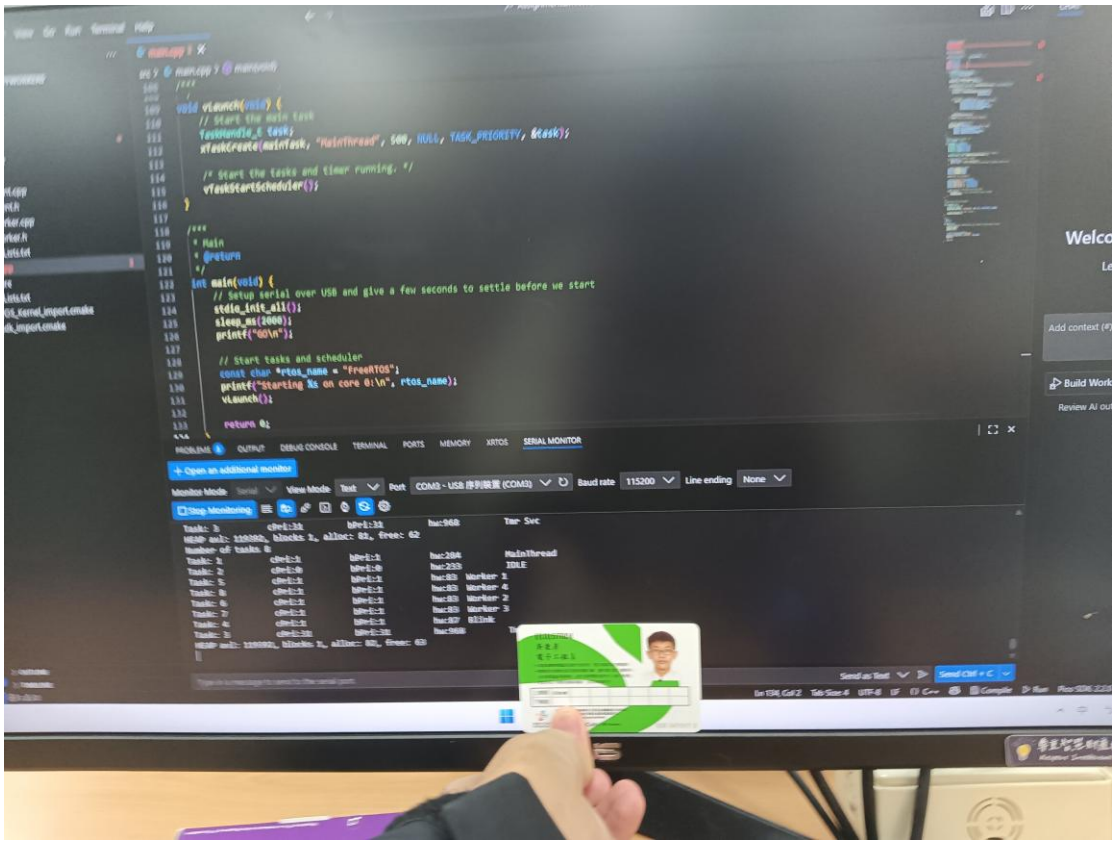
U13157004

吳秉彥

U13157004
吳秉彥
電子工程系
上學期
下學期
826 547017 5
悠遊卡 EASYCARD
MAKER PI PICO
micro SD
AUDIO
BUZZER
RGB
ESP-01
Cytron


BlinkAgent blink(LED_PAD);
BlinkAgent worker1(LED1_PAD);
BlinkAgent worker2(LED2_PAD);
BlinkAgent worker3(LED3_PAD);
BlinkAgent worker4(LED4_PAD);
BlinkAgent worker5(LED5_PAD);
BlinkAgent worker6(LED6_PAD);
BlinkAgent worker7(LED7_PAD);
printf("Main task started\n");
blink.start("Blink", TASK_PRIORITY);
worker1.start("Worker 1", TASK_PRIORITY);
worker2.start("Worker 2", TASK_PRIORITY + 1);
worker3.start("Worker 3", TASK_PRIORITY + 2);
worker4.start("Worker 4", TASK_PRIORITY + 3);
worker5.start("Worker 5", TASK_PRIORITY + 4);
worker6.start("Worker 6", TASK_PRIORITY + 5);
worker7.start("Worker 7", TASK_PRIORITY + 6);
while (true) { // Loop forever
runTimeStats();
vTaskDelay(3000);
}
}
PROBLEMS 15
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
MEMORY
XRTOS
SERIAL MONITOR
Open an additional monitor
Monitor Mode
Serial
View Mode
Text
Port
COM3 - USB 序列裝置 (COM3)
Baud rate
115200
Line ending
None
Baud rate
Stop Monitoring
Task: 6 cPri:2 bPri:2 hw:87 Worker 2
Task: 4 cPri:1 bPri:1 hw:87 Blink
Task: 5 cPri:1 bPri:1 hw:87 Worker 1
Task: 11 cPri:7 bPri:7 hw:87 Worker 7
Task: 3 cPri:31 bPri:31 hw:970 Tmr Svc
HEAP avl: 117256, blocks 1, alloc: 90, free: 66
Type in a message to send to the serial port.
Ln 102, Col 5
Tab Size: 4


U13157004
吳秉彥
電子工程
上學期
113學年
下學期
悠遊卡 EASYCARD
學生 Student
826 547017 5
MAKER PI PICO
micro SD
AUDIO
BUZZER
RUN
ESP-01
RGB
Cytron
GROVE 4
GROVE 5
GROVE 6


main.cpp
/***
void vLaunch(void) {
// Start the main task
TaskHandle_t task;
xTaskCreate(mainTask, "MainThread", 500, NULL, TASK_PRIORITY, &task);
/* Start the tasks and timer running. */
vTaskStartScheduler();
}
/***
* Main
* @return
*/
int main(void) {
// Setup serial over USB and give a few seconds to settle before we start
stdio_init_all();
sleep_ms(2000);
printf("go\n");
// Start tasks and scheduler
const char *rtos_name = "FreeRTOS";
printf("Starting %s on core 0:\n", rtos_name);
vLaunch();
return 0;
PROBLEMS 4
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
MEMORY
XRTOS
SERIAL MONITOR
+ Open an additional monitor
Monitor Mode
View Mode
Text
Port
COM3 - USB 序列裝置 (COM3)
Baud rate
115200
Line ending
None
Stop Monitoring
Tmr Svc
MainThread
IDLE
Worker 1
Worker 4
Worker 2
Worker 3
Blink
Send as Text
Send Ctrl + C

README

Chap 4 使得系統可以處理更多的任務，並且在 RP2040 上測試 FreeRTOS 的多任務處理能力。增加更多任務並觀察系統的邊界，有助於了解 RP2040 在處理大量並行任務時的性能表現

Chap 5 的重點是 **計數型信號量** 的應用，它有效地限制了同時運行的任務數量，確保了系統資源的合理分配。通過使用信號量，我們成功地同步了多個 worker 任務，並且在任務運行的過程中能夠實時檢查系統資源的使用情況。